A Seven-Coefficient Obstruction to Wagner's Forest Sum-of-Squares Conjecture

Research draft prepared for Keio

November 2026

Abstract

We give an exact obstruction to the prescribed signed sum-of-squares identity in Conjecture 1.3 of Erickson (2012), attributed there to Wagner. The graph is the eight-vertex Möbius ladder, and the two distinguished edges are opposite edges of its outer cycle. Seven coefficients of the proposed identity force two signs $T, U \in \{-1, 1\}$ to satisfy $T + 4U = 4$, an impossibility. The proof excludes every sign assignment, without a numerical tolerance or an unproved auxiliary assumption. A second seven-coefficient argument excludes even multiple polynomial squares with arbitrary real coefficients on the prescribed monomial supports. Edge deletion excludes the original identity on every complete graph K_n with $n \geq 8$. We also explain a general reduction of such sign identities to linear equations over $\mathbb{F}_2$ by working modulo 8. The same Rayleigh difference nevertheless has an explicit unrestricted monomial-times-squares representation, giving a strict separation between the prescribed supports and general square certificates. The forest negative-correlation conjecture is not refuted. The source explicitly reports a private verification for the Möbius ladder; our coefficient certificate conflicts with that report. The full finite verification is therefore included, and no historical-priority claim is made.

1 The exact previously stated conjecture

For a finite simple graph $G = (V, E)$, let

$$F_G(\boldsymbol{y}) = \sum_{X \in \mathcal{F}(G)} \boldsymbol{y}^X, \qquad \boldsymbol{y}^X = \prod_{g \in X} y_g,$$

where $\mathcal{F}(G)$ consists of *all* acyclic edge subsets, including the empty set. The associated forest Rayleigh difference at distinct edges e, f is

$$\Delta_{G,e,f} = (\partial_e F_G)(\partial_f F_G) - F_G \partial_e \partial_f F_G. \tag{1}$$

Writing $F_G = Z + y_e A + y_f B + y_e y_f C$ shows directly that $\Delta_{G,e,f} = AB - ZC$ is independent of y_e, y_f. The symbols A, B, C in this decomposition are used only in this paragraph.

The weighted forest negative-correlation conjecture asserts $\Delta_{G,e,f}(\boldsymbol{y}) \geq 0$ for positive weights. The relevant conjecture in this paper is the *specified signed polynomial identity* proposed as a sufficient explanation of that inequality. We use precisely Definition 1.2 and Conjecture 1.3 of Erickson [2].

Let $H = E \setminus \{e, f\}$ and set

$$\mathcal{S} = \{S \subseteq H : S \cup \{e, f\} \text{ is contained in a simple cycle of } G\}.$$

For $S \in \mathcal{S}$, let $\mathcal{A}(S)$ consist of forests $A \subseteq H$ such that $A \cup \{e, f\}$ has exactly one cycle C_A, with $S \cup \{e, f\} \subseteq C_A$. For each admissible pair (S, C) choose a sign $c(S, C) \in \{-1, 1\}$, shared by all forests having the same S and the same cycle. The conjectured identity is

$$\Delta_{G,e,f} = R_c(\boldsymbol{y}) := \sum_{S \in \mathcal{S}} \boldsymbol{y}^S \left(\sum_{A \in \mathcal{A}(S)} c(S, C_A) \boldsymbol{y}^{A \setminus S} \right)^2 . \tag{2}$$

The quantifier is that for *every* graph and *every* pair of distinct edges, some choice of these signs makes the identity hold. An earlier public statement is Conjecture 5.2.6 of Erickson's 2008 thesis [1].

Theorem 1. *Let M_8 be the graph with vertex set $\{0, 1, \ldots, 7\}$, outer cycle edges $\{i, i+1 \pmod 8\}$, and opposite chords $\{i, i+4\}$ for $0 \le i \le 3$. Take $e = 01$ and $f = 45$. There is no sign assignment $c(S, C) \in \{-1, 1\}$ satisfying (2). Consequently, the universal assertion in [2, Conjecture 1.3] is false as stated.*

Here and below uv denotes the unordered edge $\{u, v\}$. This is a statement about equality of polynomials, with the original graph, forest sets, cycles, exponents, and signs of the conjecture.

2 The graph and the seven coefficient equations

Order the ten unmarked edges as

$$(h_1, \ldots, h_{10}) = (04, 07, 12, 15, 23, 26, 34, 37, 56, 67). \tag{3}$$

Figure 1 displays the graph. The edge order in (3), rather than the drawing, specifies the certificate.

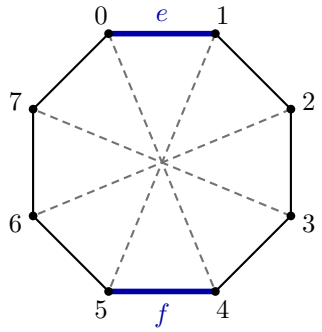


Figure 1: The eight-vertex Möbius ladder. Dashed lines are edges; their interior intersections are not vertices. The marked edges are opposite outer-cycle edges.

For a nonnegative integer mask $b < 2^{10}$, write $H_b = \{h_j : \text{bit } j-1 \text{ of } b \text{ is } 1\}$. This notation is only an exact encoding of subsets. Put

$$c_{s,b} = c(H_s, H_b \cup \{e, f\}).$$

The seven cycle masks occurring in the proof are listed in Table 1. Each listed edge set becomes a simple cycle when e and f are added.

Define the following products of signs:

$$\begin{aligned}
A &= c_{528,634}c_{528,854}, & B &= c_{528,854}c_{528,917}, & D &= c_{0,9}c_{0,854}, \\
E_* &= c_{1,9}c_{1,293}, & F_* &= c_{8,9}c_{8,634}, & G_* &= c_{32,293}c_{32,634}, \\
P &= c_{0,9}c_{0,486}, & Q &= c_{1,9}c_{1,917}, & R &= c_{128,202}c_{128,917}, \\
T &= c_{0,202}c_{0,293}, & U &= c_{8,9}c_{8,202}.
\end{aligned} \tag{4}$$

b	Indices j with $h_j \in H_b$	A cyclic vertex order
9	$1,4$	$0,1,5,4,0$
202	$2,4,7,8$	$0,1,5,4,3,7,0$
293	$1,3,6,9$	$0,1,2,6,5,4,0$
486	$2,3,6,7,8,9$	$0,1,2,6,5,4,3,7,0$
634	$2,4,5,6,7,10$	$0,1,5,4,3,2,6,7,0$
854	$2,3,5,7,9,10$	$0,1,2,3,4,5,6,7,0$
917	$1,3,5,8,9,10$	$0,1,2,3,7,6,5,4,0$

Table 1: The cycles used in the coefficient certificate.

All of these are in $\{-1,1\}$. In addition,

$$c_{528,634}c_{528,917} = AB. \tag{5}$$

The subsets indexing the signs are $H_0 = \varnothing$, $H_1 = \{h_1\}$, $H_8 = \{h_4\}$, $H_{32} = \{h_6\}$, $H_{128} = \{h_8\}$, and $H_{528} = \{h_5, h_{10}\}$. Each is contained in every cycle with which it is paired above.

For $v = (v_1, \dots, v_{10})$, write $[v]F$ for the coefficient of $\prod_j y_{h_j}^{v_j}$ in F. The following table gives five necessary coefficient equations for (2).

Row	v	$[v]\Delta_{M_8,e,f}$	$[v]R_c$
1	$(0,2,1,1,1,1,2,0,1,1)$	-2	$2A$
2	$(1,1,2,0,1,0,1,1,2,1)$	-2	$2B$
3	$(1,1,2,1,1,1,2,0,1,2)$	-2	$2D$
4	$(1,0,1,1,0,1,0,0,2,2)$	4	$2+2E_*$
5	$(1,2,1,1,1,1,2,0,1,1)$	0	$2(D+E_*+F_*+G_*)$

Table 2: Five small coefficient equations.

Row 6 has all ten exponents equal to 1. Row 7 has exponent 2 on h_3 and exponent 1 on each of the other nine edges. Their equations are

$$\begin{aligned} 74 &= 28 + 2P + 2D + 8T + 20E_* + 2Q + 20U \\ &\quad + 2F_* + 2G_* + 2R + 2AB, \end{aligned} \tag{6}$$

$$16 = 2P + 2D + 4T + 8E_* + 2Q + 4U + 2R. \tag{7}$$

Section 6 gives the complete finite procedure for reproducing every coefficient, including the separate counts in the two products defining Δ.

Proof of Theorem 1. Assume that (2) holds. The first four rows of Table 2 force

$$A = B = D = -1, \qquad E_* = 1.$$

The fifth gives $F_* + G_* = 0$, and (5) gives $AB = 1$. Substitution in (6) and (7) yields

$$P + Q + R + 4T + 10U = 13, \qquad P + Q + R + 2T + 2U = 5.$$

Subtracting shows $2T + 8U = 8$, hence

$$\boxed{T + 4U = 4.} \tag{8}$$

Since $T \in \{-1,1\}$ and U is an integer, the left side is odd and the right side is even. This is impossible. □

3 Consequences and a general parity test

Lemma 2. *If a graph G has a representation (2) for e, f, so does the graph obtained by deleting any collection of unmarked edges.*

Proof. Set the weights of the deleted edges equal to zero. The forest polynomial becomes that of the remaining graph, so the same holds for its Rayleigh difference. On the right, a term with a deleted edge in S vanishes through $\boldsymbol{y}^S$. A term with a deleted edge in $A \setminus S$ vanishes through its inner monomial. The surviving terms are exactly the original S-sets and A-sets for the remaining graph, with their unchanged signs. Isolated vertices do not affect any of these polynomials or conditions. □

Corollary 3. *For every $n \geq 8$, the complete graph K_n does not satisfy the universal signed identity (2) for all pairs of edges. More generally, any graph containing a copy of M_8 as a subgraph fails that identity for a corresponding pair of opposite rim edges.*

Proof. Delete all edges other than those of that copy of M_8, and apply Lemma 2 and Theorem 1. □

The stronger closure under minors is proved in [2, Lemma 2.1]; it is not needed for this corollary.

The short proof above can be found by a general, exact parity test. For each sign write $c_v = 1 - 2x_v$ with $x_v \in \{0, 1\}$, where v denotes an admissible pair (S, C). Let R_+ be the right side of (2) with all signs equal to $+1$. An unordered pair of distinct monomials in one square contributes $2c_v c_w$. Therefore

$$2 - 2c_v c_w = 4(x_v + x_w) - 8x_v x_w \equiv 4(x_v + x_w) \pmod 8. \tag{9}$$

If $v = w$, the product is identically 1 and contributes no variable.

Proposition 4. *For each exponent vector μ, collect all unordered cross terms of R_c that contribute to that exponent. Let $L_\mu(x)$ be the sum over $\mathbb{F}_2$ of their two sign variables, with multiplicities. If (2) holds, then*

$$[\boldsymbol{y}^\mu](R_+ - \Delta) \equiv 0 \pmod 4, \qquad L_\mu(x) = \frac{[\boldsymbol{y}^\mu](R_+ - \Delta)}{4} \pmod 2.$$

Any collection of these equations whose variable sides cancel and whose right sides sum to 1 is a complete obstruction to the signed identity.

Proof. Expand each square and apply (9) to its cross terms. All diagonal terms are sign independent. The asserted equations follow coefficient by coefficient; summing an inconsistent subset would give $0 = 1$ in $\mathbb{F}_2$. □

For the seven rows used here, the all-positive coefficients, in row order, are

$$2,\ 2,\ 2,\ 4,\ 8,\ 90,\ 24,$$

and their sum is 132. The corresponding coefficients of Δ sum to 88. The sign-variable incidences cancel over $\mathbb{F}_2$, while $(132 - 88)/4 = 11$ is odd. The direct proof leading to (8) is a hand-checkable reduction of this parity certificate. This is an exact noncancellation obstruction, not evidence obtained by testing finitely many numerical weight values.

4 Arbitrary coefficients on the prescribed supports also fail

For each S, let V_S be a real inner-product space of arbitrary dimension, and choose arbitrary vectors $v_{S,A} \in V_S$ for $A \in \mathcal{A}(S)$. Consider the larger class of representations

$$\Delta_{G,e,f} = \sum_{S \in \mathcal{S}} \boldsymbol{y}^S \left\| \sum_{A \in \mathcal{A}(S)} v_{S,A} \boldsymbol{y}^{A \setminus S} \right\|^2 . \tag{10}$$

No norm or cycle-dependence assumption is imposed. Resolving each norm into coordinates allows any number of polynomial squares per S and arbitrary real coefficients, on the prescribed monomial supports.

Lemma 5 (Forced norms and cycle dependence)**.** *If* (10) *holds, then* $\|v_{S,A}\| = 1$ *and* $v_{S,A} = v_{S,D}$*, where* $D = C_A \setminus \{e, f\}$*.*

Proof. Fix $A \in \mathcal{A}(S)$ and set all unmarked weights outside A to zero. The remaining graph U has just the cycle C_A, with every edge of $A \setminus D$ a bridge. Factoring its forest polynomial gives

$$\Delta_{U,e,f} = \boldsymbol{y}^D \prod_{d \in D} (1 + y_d) \prod_{b \in A \setminus D} (1 + y_b)^2. \tag{11}$$

Indeed, the forest polynomial of the cycle is $\prod_{g \in C_A}(1+y_g) - \boldsymbol{y}^{C_A}$, and all other edges contribute factors $1 + y_b$.

The coefficient at $2\mathbf{1}_A - \mathbf{1}_S$ in (11) is 1. Every contributing pair on the right of (10) has its two forests A', B' inside A: an outside edge cannot disappear from $\mathbf{1}_{A'} + \mathbf{1}_{B'} - \mathbf{1}_{S'}$, since $S' \subseteq A' \cap B'$. The unique-cycle condition forces $D \subseteq A' \cap B'$ and $S' \subseteq D$. The exponent therefore forces $A' = B' = A$ and $S' = S$. Thus $\|v_{S,A}\|^2 = 1$.

For $b \in A \setminus D$, the coefficient at $2\mathbf{1}_A - \mathbf{1}_S - \mathbf{1}_{\{b\}}$ is 2. The same argument leaves exactly the ordered pairs $(A, A \setminus \{b\})$ and $(A \setminus \{b\}, A)$, with $S' = S$. Hence $\langle v_{S,A}, v_{S,A\setminus\{b\}} \rangle = 1$. These unit vectors are equal. Deleting the edges of $A \setminus D$ successively proves $v_{S,A} = v_{S,D}$. □

Theorem 6. *For* $M_8, e = 01, f = 45$*, no representation* (10) *exists.*

Proof. By Lemma 5, the vectors are unit vectors depending only on S, C; denote them by $u_{S,C}$. In the earlier mask notation, put $C_b = H_b \cup \{e, f\}$ and

$$h_{s;b,d} = \frac{1 - \langle u_{H_s,C_b}, u_{H_s,C_d} \rangle}{2} \geq 0.$$

Nonnegativity follows from Cauchy–Schwarz. Use the abbreviations

symbol	a	b	c	d	t	u	v	w	z	r	s	q
subset mask	8	1	0	0	0	8	32	1	128	528	528	528
first cycle	9	9	9	202	9	9	293	9	202	634	634	854
second cycle	202	293	486	293	854	634	634	917	917	854	917	917

For $\mu \in \{0, 1, 2\}^{10}$, set $\mathrm{code}(\mu) = \sum_{j=1}^{10} \mu_j 3^{j-1}$, $d_\mu = [\mu]\Delta$, and $q_\mu = [\mu]R_+$. Seven coefficient comparisons yield

Code	d_μ	q_μ	Necessary equation
9841	78	78	$8a+8b+c+2d=0$
27094	34	38	$t=1$
27337	56	64	$u+9b+t+v=2$
28068	-2	2	$r=1$
29281	56	64	$9a+w+t+z=2$
29524	74	90	$10a+u+10b+w+t+4d+c+z+v+s=4$
35824	-2	2	$q=1$

Each right side is $(q_\mu - d_\mu)/4$. An unordered cross term $2\langle u, v\rangle = 2 - 4h$ contributes one h on the left, with its multiplicity. Section 6 and the appendix program reproduce these exact coefficients.

The first equation gives $a=b=c=d=0$. The next equations yield $t=1$ and $u+v=w+z=1$. The equation with code 29524 then forces $s=1$; also $r=q=1$. The three unit vectors indexed by $(528,634),(528,854),(528,917)$ consequently have Gram matrix

$$\begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}.$$

Their sum would have squared norm $3-6=-3$, a contradiction. □

There is also a single explicit certificate of this contradiction. Combine the seven equations in their displayed order with weights $(-2,1,-1,1,-1,1,1)$. The result is $r+s+q-15a-15b-c=3$. Writing $u_{s,b}$ for u_{H_s,C_b} gives

$$\begin{aligned} &\|u_{528,634}+u_{528,854}+u_{528,917}\|^2 \\ &\quad +15\|u_{8,9}-u_{8,202}\|^2+15\|u_{1,9}-u_{1,293}\|^2+\|u_{0,9}-u_{0,486}\|^2=-3. \end{aligned} \tag{12}$$

Every term on the left is nonnegative. This coefficient functional is a fully explicit obstruction, independent of numerical semidefinite optimization.

This rules out arbitrary real coefficients and any number of squares within the original supports. Representations using additional monomials or different outer index sets remain outside the theorem.

5 An unrestricted square certificate for the same polynomial

We now show that the larger class of monomial-times-squares representations actually succeeds for this same graph and marked edge pair. Keep all the original S-sets and A-sets, and define an explicit sign rule using the masks above:

$$\varepsilon(s,b)=\begin{cases} -1, & b=854, \\ -1, & b=634 \text{ and } h_6\in H_s, \\ -1, & b=917 \text{ and } h_8\in H_s, \\ +1, & \text{otherwise.} \end{cases} \tag{13}$$

Let $P_s=\sum_{A\in\mathcal{A}(H_s)}\varepsilon(s,b_A)\boldsymbol{y}^{A\setminus H_s}$, where $C_A=H_{b_A}\cup\{e,f\}$, and put $M=\prod_{j=1}^{10}y_{h_j}$.

Lemma 7. *For this sign rule the following exact polynomial identity holds:*

$$\sum_{H_s\in\mathcal{S}}\boldsymbol{y}^{H_s}P_s^2=\Delta_{M_8,e,f}+4M. \tag{14}$$

Proof. Expand the two sides by the finite enumeration in Section 6. There are 209 nonempty square blocks. The difference has exactly one nonzero coefficient: 4 at the all-ones exponent vector. The appendix program verifies this statement over all exponents, including those with zero coefficients. In particular, this is an identity check, not an evaluation at selected weights. □

Theorem 8. *The polynomial $\Delta_{M_8,01,45}$ is a sum of positive monomial multiples of polynomial squares, although it has no representation on the prescribed supports in (10). In particular, $\Delta_{M_8,01,45}(\boldsymbol{y}) \geq 0$ for all nonnegative edge weights.*

Proof. The block $s = 1$, namely $H_s = \{h_1\} = \{04\}$, has 149 monomials, all with coefficient $+1$. It contains

$$u = y_{h_2} y_{h_4} y_{h_6} y_{h_7}, \qquad v = y_{h_3} y_{h_5} y_{h_8} y_{h_9} y_{h_{10}}.$$

These have masks 106 and 916, respectively, and $y_{h_1} uv = M$. Put $R = P_1 - u - v$; all its 147 coefficients are $+1$. By (14),

$$\begin{aligned} \Delta_{M_8,e,f} = & \sum_{\substack{H_s \in \mathcal{S} \\ s \neq 1}} \boldsymbol{y}^{H_s} P_s^2 \\ & + y_{h_1}(u-v)^2 + y_{h_1} R^2 + 2y_{h_1}(u+v)R. \end{aligned} \tag{15}$$

Here we used $(u+v+R)^2 - 4uv = (u-v)^2 + R^2 + 2(u+v)R$. The last term has only positive coefficients. Every positive monomial $a\boldsymbol{y}^\alpha$ is a positive monomial multiple of a square; more explicitly,

$$a\boldsymbol{y}^\alpha = \boldsymbol{y}^{\alpha \bmod 2} \big(\sqrt{a}\, \boldsymbol{y}^{\lfloor \alpha/2 \rfloor}\big)^2.$$

Thus (15) is the required general certificate. The impossibility on the original supports is Theorem 6. □

The new certificate genuinely leaves the original index sets. For example, R contains y_{h_4}, so its last term includes

$$2y_{h_1} y_{h_2} y_{h_6} y_{h_7} (y_{h_4})^2.$$

The outer edge set is $\{04, 07, 26, 34\}$. Together with the marked edge 01, it has three edges incident to vertex 0; it therefore cannot be contained in a simple cycle. This outer set is not in $\mathcal{S}$. The additional term is exactly the type of freedom excluded by the original conjecture.

The constructive part concerns the stated pair $01, 45$ on M_8. It does not assert negative correlation for arbitrary marked pairs on every graph, or a general square representation for K_n.

6 Complete coefficient verification

Here is a finite procedure for checking the coefficient table directly from the definitions. For an edge set X, let $k(X)$ be the number of connected components of the spanning graph (V, X), counting isolated vertices. Then

$$X \text{ is a forest} \iff |X| = 8 - k(X).$$

Enumerate the $2^{12} = 4096$ edge subsets of M_8. Retain the forests and separate them according to the presence of e, f. If Z, A, B, C are the four coefficient polynomials of $F_G = Z + y_e A + y_f B + y_e y_f C$, the two products AB and ZC give the following separate counts.

Row	Coefficient in AB	Coefficient in ZC	Difference
1	32	34	-2
2	32	34	-2
3	6	8	-2
4	12	8	4
5	36	36	0
6	372	298	74
7	102	86	16

Table 3: Integer counts on the left side of the proposed identity.

For its right side, enumerate all forests $A \subseteq H$ such that $A \cup \{e, f\}$ has cycle rank one and its unique cycle contains both marked edges. Repeatedly remove degree-one vertices; the remaining edges are exactly that cycle C_A. For every $S \subseteq C_A \setminus \{e, f\}$, retain the term (S, A, C_A). An ordered pair $A, B \in \mathcal{A}(S)$ contributes

$$c(S, C_A)c(S, C_B) \quad \text{at exponent} \quad \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_S.$$

This enumerates every term once with the multiplicity of the square expansion. Products with identical sign variables equal 1. No sign search is required: the verifier retains these products symbolically and compares them to the displayed table.

For the seven vector-certificate rows, the separate product counts are, in their displayed order,

$$296 - 218, \quad 194 - 160, \quad 292 - 236, \quad 32 - 34, \quad 292 - 236, \quad 372 - 298, \quad 32 - 34.$$

The same expansion supplies the inner-product multiplicities.

Appendix A includes a complete Python program using only the standard library. It contains the graph and the expected table; it reads no auxiliary data. The entire coefficient calculation uses integer arithmetic. The full text of this program is included in the single `.tex` source as well as in the PDF.

7 Literature audit and exact scope

Earlier formulation. The signed formula is Conjecture 5.2.6 in Erickson's 2008 thesis [1], is present in the 2010 arXiv manuscript, and appears as Conjecture 1.3 in the 2012 journal paper [2]. The latter explicitly requires that $A \cup \{e, f\}$ contain no other cycle. The same source proves the formula for series-parallel graphs. Thus this paper addresses a previously stated, named assertion rather than a newly chosen surrogate condition.

A conflicting verification report. Immediately after its Conjecture 1.3, the 2012 paper reports that the formula was verified for K_7, the cube, and the Möbius ladder on eight vertices, attributing the verification to private communication. This includes exactly the graph of Theorem 1. Our explicit coefficient calculation is incompatible with that report for $e = 01, f = 45$. We do not have the underlying claimed sign certificate and do not infer a cause for the discrepancy. The printed conjecture, the graph, and the coefficient arithmetic can be checked without that certificate. The conflict is a reason to expose the full proof and seek independent review, not to omit the report.

What the obstruction does not establish. The result does not produce positive weights at which $\Delta_{G,e,f} < 0$. It does not disprove the Kahn–Grimmett–Winkler forest negative-correlation conjecture [3]. Indeed, Theorem 8 explicitly constructs a general square certificate for the same marked graph. The broader universal sum-of-squares question is restated in [4, Conjecture 1.3]; the matching conjecture number in that paper must not be confused with Erickson's more specific Conjecture 1.3.

Priority and review. The mathematical conclusion is the impossibility of the exact signed identity on an explicit graph. A search of the cited sources, their publicly accessible versions, and subsequent literature did not identify the same seven-coefficient obstruction. This search does not prove historical priority or exclude unpublished earlier work. This document is an independently reproducible research draft, not a claim of journal acceptance, community validation, or resolution of every surrounding open problem.

References

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[3] G. R. Grimmett and S. N. Winkler, Negative association in uniform forests and connected graphs, *Random Structures & Algorithms* **24** (2004), 444–460. https://doi.org/10.1002/rsa.20012.

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A Standalone exact verifier

Save the following block as `verify_mobius8.py` and run it with Python 3. It regenerates both sets of seven coefficient equations, checks their full symbolic terms, checks both final contradictions, and verifies every coefficient of the unrestricted square certificate. It writes JSON records of the counted terms beside the script. No floating-point calculation or external package is used.

```
from pathlib import Path
import json
from collections import Counter,defaultdict

OUT=Path(__file__).resolve().parent
n=8
H=[(0,4),(0,7),(1,2),(1,5),(2,3),(2,6),(3,4),(3,7),(5,6),(6,7)]
m=len(H)
e=(0,1); f=(4,5); E=H+[e,f]
assert len(set(E))==len(E)==12
assert all(0<=u<v<n for u,v in E)
assert all(sum(v in edge for edge in E)==3 for v in range(n))
assert set(E)==set(tuple(sorted((i,(i+1)%8))) for i in range(8))|set((i,i+4) for i in range(4))

def coords(mask):return tuple(int(bool(mask&(1<<j))) for j in range(m))
def decode(code):
    ans=[]
    for _ in range(m):ans.append(code%3);code//=3
    assert code==0
    return tuple(ans)
def graph(mask):
    return [edge for j,edge in enumerate(E) if mask>>j&1]
def components(es):
    adjacency=[[] for _ in range(n)]
    for u,v in es:adjacency[u].append(v);adjacency[v].append(u)
    seen=set();count=0
    for v in range(n):
        if v in seen:continue
        count+=1;stack=[v];seen.add(v)
        while stack:
            a=stack.pop()
            for b in adjacency[a]:
                if b not in seen:seen.add(b);stack.append(b)
    return count
def is_forest(es):return len(es)==n-components(es)
def unique_cycle(es):
    if len(es)-n+components(es)!=1:return None
    core=set(es)
    # Leaf pruning computes the unique 2-core; this is the simple cycle.
    while True:
        degree=[sum(v in edge for edge in core) for v in range(n)]
        leaves={v for v,d in enumerate(degree) if d==1}
        if not leaves:break
        core={edge for edge in core if not leaves.intersection(edge)}
    assert core and all(sum(v in edge for edge in core) in (0,2) for v in range(n))
    return core

# Full forest polynomial, separated by powers of the two marked variables.
parts=[[] for _ in range(4)]
for mask in range(1<<len(E)):
    g=graph(mask)
    if is_forest(g):
        marks=(mask>>m)&3
        parts[marks].append(coords(mask))
# Delta=G_e^f G_f^e-G^{ef}G_{ef}; enumerate ordered pairs directly.
want=[9841,27094,27337,28068,28069,29281,29524,29533,35824,47758,52768]
target={decode(c):c for c in want}
D=Counter();positive=Counter();negative=Counter()
for left,right,sign in [(parts[1],parts[2],1),(parts[0],parts[3],-1)]:
    for a in left:
        for b in right:
```

```
            key=tuple(x+y for x,y in zip(a,b))
            if key in target:
                D[target[key]]+=sign
                (positive if sign==1 else negative)[target[key]]+=1

# Construct A(S) directly from leaf pruning of A union {e,f}.
# No DSU, path finder, ternary exponent arithmetic,
# or precomputed C++ list is used.
groups=defaultdict(list)
for amask in range(1<<m):
    A=graph(amask)
    if not is_forest(A):continue
    augmented=graph(amask|(3<<m))
    C=unique_cycle(augmented)
    if C is None:continue
    if e not in C or f not in C:continue
    cmask=sum(1<<j for j,edge in enumerate(H) if edge in C)
    for smask in range(1<<m):
        if smask&~cmask:continue
        groups[smask].append((coords(amask),cmask))

T={code:Counter() for code in want}
for smask,As in groups.items():
    s=coords(smask)
    for a,ca in As:
        for b,cb in As:
            ex=tuple(x+y-z for x,y,z in zip(a,b,s))
            if ex not in target:continue
            va=(smask,ca);vb=(smask,cb)
            # An ordered pair contributes one; v*v=1 exactly.
            term=() if va==vb else tuple(sorted((va,vb)))
            T[target[ex]][term]+=1

def show_var(v):return 'c_%d_%d'%v
records=[]
for code in want:
    terms=T[code]
    print(code,'exponents',decode(code),'Delta=',D[code],'; RHS=',
      ' + '.join(str(k) if not v else str(k)+'*'+'*'.join(show_var(w) for w in v)
                 for v,k in sorted(terms.items())))
    records.append({'code':code,'exponents':decode(code),'delta':D[code],'positive_count':positive[
     code],'negative_count':negative[code],
      'terms':[{'coefficient':coeff,'variables':pair} for pair,coeff in sorted(terms.items())]})
# These products define the short human-readable contradiction.
products={
'A':((528,634),(528,854)),
'B':((528,854),(528,917)),
'D':((0,9),(0,854)),
'E':((1,9),(1,293)),
'F':((8,9),(8,634)),
'G':((32,293),(32,634)),
'P':((0,9),(0,486)),
'Q':((1,9),(1,917)),
'R':((128,202),(128,917)),
'T':((0,202),(0,293)),
'U':((8,9),(8,202)),
'AB':((528,634),(528,917)),
'1':(),
}
expected={
28068:(-2,{'A':2}),
35824:(-2,{'B':2}),
47758:(-2,{'D':2}),
52768:(4,{'1':2,'E':2}),
28069:(0,{'D':2,'E':2,'F':2,'G':2}),
29524:(74,{'1':28,'P':2,'D':2,'T':8,'E':20,'Q':2,'U':20,'F':2,'G':2,'R':2,'AB':2}),
29533:(16,{'P':2,'D':2,'T':4,'E':8,'Q':2,'U':4,'R':2}),
}
for code,(delta,terms) in expected.items():
    assert D[code]==delta
    assert T[code]==Counter({tuple(sorted(products[label])):coefficient for label,coefficient in
     terms.items()})
```

```
    print('LHS counts',code,positive[code],'-',negative[code],'=',D[code])
# First four equations force A=B=D=-1, E=1. Fifth gives F+G=0.
# Sixth minus seventh therefore forces 58=42+4T+16U.
assert 74-16==58 and 28+12+2==42
assert all(t+4*u!=4 for t in (-1,1) for u in (-1,1))
print('PASS: independent exact coefficient enumeration proves no Wagner sign assignment exists.')
(OUT/'mobius8_direct_coefficients.json').write_text(json.dumps({'edges':H,'e':e,'f':f,'rows':records},indent=2)+'\n')

# Transform the independently enumerated exact Gram coefficients.
# Equal-cycle pairs have inner product one by the unit-norm assumption.
# Other Gram entries are 1-2*h, where h is nonnegative.
names={
 'a':(8,9,202),'b':(1,9,293),'c':(0,9,486),'d':(0,202,293),
 't':(0,9,854),'u':(8,9,634),'v':(32,293,634),
 'w':(1,9,917),'z':(128,202,917),
 'r':(528,634,854),'s':(528,634,917),'q':(528,854,917),
}
reverse={v:k for k,v in names.items()}
expected={
9841:(78,78,{'a':8,'b':8,'c':1,'d':2},0),
27094:(34,38,{'t':1},1),
27337:(56,64,{'u':1,'b':9,'t':1,'v':1},2),
28068:(-2,2,{'r':1},1),
29281:(56,64,{'a':9,'w':1,'t':1,'z':1},2),
29524:(74,90,{'a':10,'u':1,'b':10,'w':1,'t':1,'d':4,'c':1,'z':1,'v':1,'s':1},4),
35824:(-2,2,{'q':1},1),
}
gram_rows=[]
for code in expected:
    terms=T[code];all_plus=sum(terms.values());h_terms=Counter()
    for pair,coefficient in terms.items():
        if not pair:continue
        ((s1,c1),(s2,c2))=pair
        assert s1==s2 and c1<c2 and coefficient%2==0
        h_terms[reverse[(s1,c1,c2)]]+=coefficient//2
    assert (all_plus-D[code])%4==0
    rhs=(all_plus-D[code])//4
    assert (D[code],all_plus,dict(h_terms),rhs)==expected[code]
    print('GRAM',code,'Delta',positive[code],'-',negative[code],'=',D[code],
          '; all_plus',all_plus,'; h equation',dict(h_terms),'=',rhs)
    gram_rows.append({'code':code,'exponents':decode(code),'delta':D[code],
      'positive_count':positive[code],'negative_count':negative[code],
      'all_plus':all_plus,'h_coefficients':dict(h_terms),'rhs':rhs,
      'full_gram_terms':[{'coefficient':coeff,'variables':pair} for pair,coeff in sorted(terms.items())]})
# Symbolic proof: nonnegativity and row 9841 force a=b=c=d=0.
# Rows 27094, 27337 and 29281 then force t=1, u+v=w+z=1.
# Row 29524 forces s=1; rows 28068,35824 force r=q=1.
# Thus the three S=528 unit vectors have Gram matrix K below.
K=[[1,-1,-1],[-1,1,-1],[-1,-1,1]]
z=[1,1,1]
value=sum(z[i]*K[i][j]*z[j] for i in range(3) for j in range(3))
assert value==-3
# A direct separating certificate: the signed sum of seven equations.
weights=[-2,1,-1,1,-1,1,1]
combo=Counter();combo_rhs=0
for weight,row in zip(weights,gram_rows):
    combo_rhs+=weight*row['rhs']
    for name,coeff in row['h_coefficients'].items():combo[name]+=weight*coeff
combo={name:coeff for name,coeff in combo.items() if coeff}
assert combo=={'a':-15,'b':-15,'c':-1,'r':1,'s':1,'q':1}
assert combo_rhs==3
# 9-4*(r+s+q) is the squared norm of the sum of the three
# S=528 vectors. Also 4*h=||u-v||^2. Consequently the equations
# force the sum of FOUR nonnegative squared norms to equal -3.
direct_certificate={
 'row_weights':weights,'h_linear_combination':combo,'rhs':combo_rhs,
 'nonnegative_squared_norms':[
  {'coefficient':1,'vectors':[[528,634],[528,854],[528,917]],'signs':[1,1,1]},
  {'coefficient':15,'vectors':[[8,9],[8,202]],'signs':[1,-1]},
  {'coefficient':15,'vectors':[[1,9],[1,293]],'signs':[1,-1]},
```

```
  {'coefficient':1,'vectors':[[0,9],[0,486]],'signs':[1,-1]}],
 'impossible_sum':-3,
}
print('DIRECT CERTIFICATE',combo,'=',combo_rhs,'; four squared norms would sum to -3')
out={'graph':H,'e':e,'f':f,'rows':gram_rows,'h_definitions':names,
     'forbidden_gram':K,'negative_direction':z,'quadratic_form_value':value,
     'direct_four_square_certificate':direct_certificate,
     'status':'All seven exact equations independently reproduced by DFS plus leaf pruning; unit Gram
      contradiction verified.'}
(OUT/'mobius8_unit_gram.independent_leaf_proof.json').write_text(json.dumps(out,indent=2)+'\n')
print('PASS: independent exact enumeration confirms the seven-row UNIT-VECTOR Gram obstruction.')

# Full-coefficient verification of the unrestricted SOS repair.
# Reuse only the graph data classified above by DFS and leaf pruning.
def binary(a):
    return sum(1<<j for j,x in enumerate(a) if x)
def ternary(mask):
    return sum(3**j for j in range(m) if mask>>j&1)
def eps(s,b):
    negative=(b==854 or (b==634 and s&32)
              or (b==917 and s&128))
    return -1 if negative else 1
pc=[[ternary(binary(a)) for a in part] for part in parts]
full_delta=Counter()
for a in pc[1]:
    for b in pc[2]:full_delta[a+b]+=1
for a in pc[0]:
    for b in pc[3]:full_delta[a+b]-=1
polys={s:{binary(a)^s:eps(s,b) for a,b in terms}
       for s,terms in groups.items()}
def square(s,p):
    ans=Counter()
    for a,ca in p.items():
        for b,cb in p.items():
            ans[ternary(s)+ternary(a)+ternary(b)]+=ca*cb
    return ans
full_q=Counter()
for s,p in polys.items():full_q.update(square(s,p))
allones=sum(3**j for j in range(m))
residual={k:full_q[k]-full_delta[k]
          for k in full_q.keys()|full_delta.keys()
          if full_q[k]!=full_delta[k]}
assert residual=={allones:4}
u,v=106,916
p=polys[1]
assert p[u]==p[v]==1 and all(c==1 for c in p.values())
r={a:c for a,c in p.items() if a not in (u,v)}
assert len(p)==149 and len(r)==147 and len(polys)==209
assert u&v==0 and u|v==1022
repaired=full_q.copy();repaired.subtract(square(1,p))
repaired.update(square(1,{u:1,v:-1}))
repaired.update(square(1,r))
positive=Counter()
for a,c in r.items():
    positive[ternary(1)+ternary(u)+ternary(a)]+=2*c
    positive[ternary(1)+ternary(v)+ternary(a)]+=2*c
assert len(positive)==294 and all(c>0 for c in positive.values())
repaired.update(positive)
assert all(repaired[k]==full_delta[k]
           for k in repaired.keys()|full_delta.keys())
assert sum(c!=0 for c in full_delta.values())==16838
print('PASS: all coefficients of the unrestricted SOS match Delta.')
print('Nonzero Delta coefficients: 16838; repaired square blocks: 504.')
```